

Matriks Varians-Covarians dan Matriks Korelasi

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Matriks Varians-Covarians

- Matriks berbentuk persegi
- Diagonal Utama: Berisi nilai Varians (sebaran data untuk tiap peubah itu sendiri).
- Off-Diagonal: Berisi nilai Kovarians (hubungan kedekatan antara dua peubah yang berbeda).
- Simetris : $s_{ij} = s_{ji}$
- Disimbolkan dengan (Σ) untuk Populasi ATAU (**S**) untuk Sampel.

Matriks Varians-Covarians

- Struktur Matriks Varians-Kovarians jika mempunyai p variable:

$$S = \begin{bmatrix} s_1^2 & s_{12} & s_{13} & \dots & s_{1p} \\ s_{21} & s_2^2 & s_{23} & \dots & s_{2p} \\ s_{31} & s_{32} & s_3^2 & \dots & s_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & s_{p3} & \dots & s_p^2 \end{bmatrix}$$

- Contoh jika mempunyai 3 variable, (X1, X2, dan X3) maka struktur matriksnya menjadi :

$$S = \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix}$$

- Varians (s^2) : $s_x^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$
- Kovarians (s_{xy}) : $s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1}$

Matriks Varians-Covarians

- Contoh :

No (i)	X1	X2	X3
1	4	65	7
2	6	80	4
3	3	60	8
4	5	75	5
5	7	90	3
6	2	50	9
7	5	70	6
8	8	95	2
9	4	70	7
10	6	85	4
Sum	50	740	55
Mean	5	74	5.5

Matriks Varians-Covarians

- Cari

$$d_{ij} = (x_{ij} - \bar{x}_j)$$

No (i)	X1	X2	X3	d1=(X1-5)	d2=(X2-74)	d3=(X3-5.5)
1	4	65	7	-1	-9	1.5
2	6	80	4	1	6	-1.5
3	3	60	8	-2	-14	2.5
4	5	75	5	0	1	-0.5
5	7	90	3	2	16	-2.5
6	2	50	9	-3	-24	3.5
7	5	70	6	0	-4	0.5
8	8	95	2	3	21	-3.5
9	4	70	7	-1	-4	1.5
10	6	85	4	1	11	-1.5
Sum	50	740	55			
Mean	5	74	5.5			

Matriks Varians-Covarians

- Cari
Sum of Squares
Dan
Cross Product

d1=(X1-5)	d2=(X2-74)	d3=(X3-5.5)	d_1^2	d_2^2	d_3^2	d1.d2	d1.d3	d2.d3
-1	-9	1.5	1	81	2.25	9	-1.5	-13.5
1	6	-1.5	1	36	2.25	6	-1.5	-9
-2	-14	2.5	4	196	6.25	28	-5	-35
0	1	-0.5	0	1	0.25	0	0	-0.5
2	16	-2.5	4	256	6.25	32	-5	-40
-3	-24	3.5	9	576	12.25	72	-10.5	-84
0	-4	0.5	0	16	0.25	0	0	-2
3	21	-3.5	9	441	12.25	63	-10.5	-73.5
-1	-4	1.5	1	16	2.25	4	-1.5	-6
1	11	-1.5	1	121	2.25	11	-1.5	-16.5
SUM			30	1740	46.5	225	-37	-280
SUM/(n-1)			3.333	193.333	5.167	25.000	-4.111	-31.111

Matriks Varians-Covarians

- Susun
Matriks S

	d_1^2	d_2^2	d_3^2	d1.d2	d1.d3	d2.d3
SUM/(n-1)	3.333	193.333	5.167	25.000	-4.111	-31.111

$$\text{Matriks Varians - Covarians} = S = \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix}$$

$$\text{Matriks Varians - Covarians} = S = \begin{bmatrix} 3.33 & 25.00 & -4.111 \\ 25.00 & 193.33 & -31.111 \\ -4.111 & -31.111 & 5.167 \end{bmatrix}$$

Matriks Korelasi

- Bentuk standarisasi dari Matriks Varian-Kovarian (S).
- Menghilangkan efek skala/satuan.
- Matriks berbentuk persegi
- Diagonal Utama: Selalu bernilai **1** (karena korelasi peubah dengan dirinya sendiri pasti sempurna).
- Off-Diagonal: Berisi nilai r_{ij} (koefisien korelasi Pearson antara peubah i dan j).
- Simetris : $r_{ij} = r_{ji}$
- Disimbolkan dengan (**P**) untuk Populasi ATAU (**R**) untuk Sampel.

Matriks Korelasi

- Struktur Matriks Korelasi jika mempunyai p variable:

$$R = \begin{bmatrix} 1 & r_{12} & r_{13} & \dots & r_{1p} \\ r_{21} & 1 & r_{23} & \dots & r_{2p} \\ r_{31} & r_{32} & 1 & \dots & r_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & r_{p3} & \dots & 1 \end{bmatrix}$$

- Contoh jika mempunyai 3 variable, (X1, X2, dan X3) maka struktur matriksnya menjadi :

$$R = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{bmatrix}$$

- Korelasi Pearson = $r_{ij} = \frac{\sum (x_i - \bar{x})(x_j - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (x_j - \bar{y})^2}}$ atau $r_{ij} = \frac{SP_{ij}}{\sqrt{SS_i \cdot SS_j}}$

Matriks Korelasi

- Contoh :

No (i)	X1	X2	X3
1	4	65	7
2	6	80	4
3	3	60	8
4	5	75	5
5	7	90	3
6	2	50	9
7	5	70	6
8	8	95	2
9	4	70	7
10	6	85	4
Sum	50	740	55
Mean	5	74	5.5

Matriks Korelasi

- Lakukan langkah yang lama dengan perhitungan Varians-covarians

$d1=(X1-5)$	$d2=(X2-74)$	$d3=(X3-5.5)$	d_1^2	d_2^2	d_3^2	d1.d2	d1.d3	d2.d3
-1	-9	1.5	1	81	2.25	9	-1.5	-13.5
1	6	-1.5	1	36	2.25	6	-1.5	-9
-2	-14	2.5	4	196	6.25	28	-5	-35
0	1	-0.5	0	1	0.25	0	0	-0.5
2	16	-2.5	4	256	6.25	32	-5	-40
-3	-24	3.5	9	576	12.25	72	-10.5	-84
0	-4	0.5	0	16	0.25	0	0	-2
3	21	-3.5	9	441	12.25	63	-10.5	-73.5
-1	-4	1.5	1	16	2.25	4	-1.5	-6
1	11	-1.5	1	121	2.25	11	-1.5	-16.5
SUM			30	1740	46.5	225	-37	-280
SUM/(n-1)			3.333	193.333	5.167	25.000	-4.111	-31.111

Matriks Korelasi

- Hitung off diagonal matriks korelasinya dengan Pearson

$$r_{ij} = \frac{\sum (x_i - \bar{x})(x_j - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (x_j - \bar{y})^2}} \text{ atau } r_{ij} = \frac{SP_{ij}}{\sqrt{SS_i \cdot SS_j}}$$

$$r_{12} = \frac{\sum d1.d2}{\sqrt{\sum d_1^2 \times \sum d_2^2}} = \frac{225}{\sqrt{30 \times 1740}} = \frac{225}{\sqrt{52200}} = \frac{225}{228.4732} = 0.9848$$

$$r_{13} = \frac{\sum d1.d3}{\sqrt{\sum d_1^2 \times \sum d_3^2}} = \frac{-37}{\sqrt{30 \times 46.5}} = \frac{-37}{\sqrt{1395}} = \frac{-37}{37.35} = -0.9906$$

$$r_{23} = \frac{\sum d2.d3}{\sqrt{\sum d_2^2 \times \sum d_3^2}} = \frac{-280}{\sqrt{1740 \times 46.5}} = \frac{-280}{\sqrt{80910}} = \frac{-280}{284.45} = -0.9844$$

Matriks Korelasi

- Susun Matriks Korelasinya

$$R = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{bmatrix}$$

$$\text{Matriks Korelasi} = R = \begin{bmatrix} 1 & 0.9848 & -0.9906 \\ 0.9848 & 1 & -0.9844 \\ -0.9906 & -0.9844 & 1 \end{bmatrix}$$

Implementasi di R

Input Data

```
data_latihan <- data.frame( x1 = c(4, 6, 3, 5, 7, 2, 5, 8, 4, 6), x2 = c(65, 80, 60, 75, 90, 50, 70, 95, 70, 85), x3 = c(7, 4, 8, 5, 3, 9, 6, 2, 7, 4))
```

Matriks Varian-Kovarian (S)

```
S <- cov(data_latihan)
round(S, 3)
```

Matriks Korelasi (R)

```
R <- cor(data_latihan)
round(R, 4)
```

Matriks Korelasi (R) Jika dibuat dari Matriks Varians-Covarians

```
R_fromCov <- cov2cor(S)
round(R_fromCov, 4)
```

```
> S <- cov(data_latihan)
> round(S, 3)
      x1      x2      x3
x1  3.333  25.000 -4.111
x2  25.000 193.333 -31.111
x3  -4.111 -31.111  5.167
```

```
> R <- cor(data_latihan)
> round(R, 4)
      x1      x2      x3
x1  1.0000  0.9848 -0.9906
x2  0.9848  1.0000 -0.9844
x3 -0.9906 -0.9844  1.0000
```

```
> R_fromCov <- cov2cor(S)
> round(R_fromCov, 4)
      x1      x2      x3
x1  1.0000  0.9848 -0.9906
x2  0.9848  1.0000 -0.9844
x3 -0.9906 -0.9844  1.0000
```

Terimakasih